Reg. No.															

B.Tech. / M.Tech (Integrated) DEGREE EXAMINATION, JULY 2023
Second Semester

21MAB102T – ADVANCED CALCULUS AND COMPLEX ANALYSIS
(For the candidates admitted from the academic year 2022-2023)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part – B** and **Part – C** should be answered in answer booklet.

Time: 3 Hours | Max. Marks: 75

PART – A (20 × 1 = 20Marks)
Answer **ALL** Questions

	Marks	BL	CO	PO
1. Evaluation of $\int_0^{\pi}\int_0^{\pi} d\theta d\phi$ is	1	1	1	1
(A) 1 (B) 0 (C) $\pi/2$ (D) π^2				
2. The area of an ellipse is	1	1	1	1
(A) πr^2 (B) $\pi a^2 b$ (C) πab (D) πab^2				
3. If R is the region bounded by $x=0$, $y=0$, $x+y=1$ then $\iint_R dxdy$ is equal to	1	2	1	1
(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$				
4. Area of the double integral in polar coordinates is equal to	1	2	1	2
(A) $\iint_R drd\theta$ (B) $\iint_R r^2 drd\theta$ (C) $\iint_R (r+1) drd\theta$ (D) $\iint_R rdrd\theta$				
5. If r is the position vector of the point (x,y,z) with respect to the origin then $\nabla \cdot \vec{r}$	1	2	2	2
(A) 2 (B) 3 (C) 0 (D) 1				
6. The unit vector normal to the surface $x^2+y^2-z^2=1$ at $(1,1,1)$ is	1	2	2	2
(A) $\vec{i}+\vec{j}-\vec{k}/\sqrt{3}$ (B) $2\vec{i}+2\vec{j}-2\vec{k}/\sqrt{2}$ (C) $\frac{3\vec{i}+3\vec{j}-3\vec{k}}{2\sqrt{3}}$ (D) $\frac{\vec{i}+\vec{j}-\vec{k}}{3\sqrt{2}}$				

THE HELPERS

7. If ϕ is a scalar point function then curl(gradϕ) is 1 2 2 1

(A) Solenoidal (B) Irrotational
(C) Constant (D) 0

8. If the value of $\int_A^B \vec{F}.d\vec{r}$ does not depend on the curve C but only on terminal points A and B then $\vec{F}$ is called 1 2 2 1

(A) Solenoidal vector (B) Irrotational vector
(C) Conservative vector (D) Neither conservative nor irrotational

9. $L\left[e^{3t}\right]=$_____ 1 1 3 2

(A) $\frac{1}{s+3}$ (B) $\frac{1}{s-3}$
(C) $\frac{3}{s+3}$ (D) $\frac{s}{s-3}$

10. An example of a function for which the laplace transforms does not exits is 1 1 3 2

(A) $f(t)=t^2$ (B) $f(t)=\tan t$
(C) $f(t)=\sin t$ (D) $f(t)=e^{-at}$

11. Inverse laplace transform of $\frac{1}{s^2-a^2}$ is 1 1 3 1

(A) $\frac{\sin at}{a}$ (B) $\frac{\sinh at}{a}$
(C) $\sin at$ (D) $\sinh at$

12. Inverse laplace transform of $\frac{1}{s^2}$ is 1 1 3 1

(A) $t/2$ (B) $t^2/2$
(C) t (D) t^2

13. The function $f(z)=u+iv$ is analytic if 1 2 4 1

(A) $u_x=v_y; u_y=-v_x$ (B) $u_x=-v_y; u_y=v_x$
(C) $u_x+v_y=0; u_y-v_x=0$ (D) $u_x=v_y; u_y=v_x$

14. If $u+iv$ is analytic, then the curves $u=c_1$ and $v=c_2$. 1 2 4 2

(A) are parallel (B) Coincides
(C) Intersect each other (D) Cut orthogonally

15. The transformation W=CZ where C is a real constant is known as 1 2 4 2

(A) Rotation (B) Reflection
(C) Magnification (D) Magnification and rotation

16. An analytic function with constant modulus is 1 2 4 1

(A) Zero (B) Analytic
(C) Constant (D) Harmonic

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17.	If $f(z)$ is analytic inside and on C, the value of $\int_c \frac{f(z)}{z-a}dz$ where C is the simple closed curve and a is any point within C is (A) $f(a)$ (B) $2\pi i f(a)$ (C) $2\pi i\, f'(a)$ (D) 0	1	1	5	2
18.	If $f(z)$ is not analytic at $z=z_0$ and there exists a neighborhood of $z=z_0$ containing no other singularity, then (A) $z=z_0$ is isolated singularity of f(z) (B) $z=z_0$ is zero of f(z) (C) $z=z_0$ is non zero of f(z) (D) $z=z_0$ is non isolated singularity of f(z)	1	1	5	2
19.	If $f(z)=\frac{\sin z}{z}$, then (A) z=0 is a simple pole (B) z=0 is a pole of order 2 (C) z=0 is a removable singularity (D) z=0 is a zero of f(z)	1	1	5	2
20.	$f(z)=\frac{z+2}{(z-1)^2(z-2)}$ has (A) Only poles (B) A simple pole at z=1/2 (C) Essential singularity (D) No poles	1	1	5	1

PART – B (5 × 8 = 40 Marks)

Answer **ALL** Questions

		Marks	BL	CO	PO
21.a.	Change the order of integration and evaluation $\int_0^{4a}\int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dydx$.	8	3	1	3
	(OR)				
21.b.	Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} dzdydx$.	8	3	1	3
22.a.	Find a, b, c so that the vector $\overrightarrow{F}=(x+2y+az)\vec{i}+(bx-3y-z)\vec{j}+(4x+cy+2z)\vec{k}$ is irrotational.	8	3	2	3
	(OR)				
22.b.	Show that $\overrightarrow{F}=(2xy+z^3)\vec{i}-x^2\vec{j}+3xz^2\vec{k}$ is a conservative vector field. Find the scalar potential also.	8	3	2	2

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23.a.	Find the laplace transform of the rectangular wave function given by $f(t)=\begin{cases}1 & ;\ 0<t<b\\ -1 & ;\ b<t<2b\end{cases}$	8	3	3	3
	(OR)				
23.b.	Find $L^{-1}\left[\log\left\{\frac{s(s+1)}{s^2+1}\right\}\right]$.	8	3	3	2
24.a.	Find the bilinear map which maps the points $z=1,i,-1$ on to the point $\omega=i,0,-i$.	8	3	4	3
	(OR)				
24.b.	Find the analytic function $f(z)=u+iv$ where $u-v=e^x(\cos y-\sin y)$.	8	3	4	3
25.a.	If $f(z)=\frac{1}{(z-1)(z-2)}$ expand as Laurent's series about $z=0$ valid in (i) $\lvert z\rvert<1$ (ii) $1<\lvert z\rvert<2$	8	4	5	3
	(OR)				
25.b.	Evaluate $\oint_c \frac{\sin\pi z^2+\cos\pi z^2}{z+z^2}dz$ where c is the circle $\lvert z\rvert=2$.	8	4	5	2

PART – C (1 × 15 = 15 Marks)

Answer **ANY ONE** Questions

		Marks	BL	CO	PO
27.	Evaluate $\int_0^{2\pi}\frac{d\theta}{13+5\sin\theta}$ using contour integration.	15	4	5	3
28.	Verify divergence theorem for $\vec{F}=\left(x^2-yz\right)\vec{i}+\left(y^2-zx\right)\vec{j}+\left(z^2-xy\right)\vec{k}$ taken over the rectangular parallelopiped $0\le x\le a;0\le y\le b;0\le z\le c$.	15	4	2	3

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